Qwen

Verifiable Autonomy: A Monotone Capability Lattice for Neurorights-Architected Defense Against Cognitive Intrusion

The development of Aegis-Shield represents a significant advancement in the field of AI-mediated security, aiming to create a comprehensive defense framework that protects user accounts and brain-identity patterns from unauthorized access by both criminal actors and state entities often referred to as "LEOs" . This research report provides a deep analysis of the proposed system, dissecting its three core pillars: the formal verification of a monotone capability lattice, the integration of semantic weapon detection for intent recognition, and the implementation of a cryptographic governance pipeline for policy enforcement. The objective is to produce a detailed and insightful examination of how these components interlock to form a system grounded in the principles of cognitive liberty and neurorights, while providing practical guidance for its implementation and future development. The analysis draws upon the provided research materials, conversation history, and supporting learnings to construct a holistic understanding of the Aegis-Shield architecture, its theoretical foundations, and its practical application.

The Monotone Capability Lattice: A Formal Guarantor of Cognitive Liberty

The foundational principle of Aegis-Shield is the establishment of a formal, mathematically verifiable guarantee that no external entity can ever revoke a user's granted capabilities without explicit, reversible consent . This concept is realized through the implementation of a monotone capability lattice, a structure derived from order theory and extremal set theory where every state transition is strictly upward, meaning capabilities can only be added, not removed [1] [2]. This design directly translates the philosophical right of cognitive liberty—the freedom from unauthorized interference in one's mental processes—into a concrete, unbreakable program property [46] [163]. The system's state is defined as a pair $s=(U_s,G_s)$, where U_s is the set of user capabilities and

G_s is the set of governance capabilities . States are ordered by inclusion, such that $s_1 \leq s_2$ if and only if $U_{s_1} \subseteq U_{s_2}$ and $G_{s_1} \subseteq G_{s_2}$. The core invariant of the system is that any transition $s \to s'$ must be monotone, satisfying $s \leq s'$.

This invariant is not merely a conceptual guideline but a critical proof obligation enforced at multiple levels of the system. At the highest level of abstraction, it is expressed as an invariance formula: $\forall s_0, s_1, c.\, \text{Reach}(s_0, s_1) \wedge c \in U_{s_0} \Rightarrow c \in U_{s_1}$. This states that if a capability `c` is present in an initial state s_0, it must remain present in any reachable subsequent state s_1. This temporal property can be encoded as a Computation Tree Logic (CTL) formula, $AG(\text{UserCap}(c) \Rightarrow AX\, \text{UserCap}(c))$, which asserts that in all paths from any state, if a user capability holds now, it will hold in the next state . While full-scale model checking of a complex AI system remains a challenging endeavor [176], the invariant is enforced more pragmatically at the level of the guardian service and the CI/CD pipeline. In the proposed Rust implementation, the `CapabilityState` is represented as a struct containing two `BTreeSet`s for user and governance capabilities . A method `leq` determines if one state is less than or equal to another by checking subset inclusion on both capability sets . The `CapabilityTransition` struct captures a change from a `from` state to a `to` state, and its `is_monotone` method simply calls `from.leq(&to)` . The most critical line of code is the `assert!(tr.is_monotone(), "non-monotone transition forbidden by Aegis-Shield");` statement within the `decide` function of the `AegisGuardian` . This assertion acts as a hard runtime check, immediately halting execution if a policy or filter attempt were to generate a state transition that violates the monotonicity invariant. This approach aligns with modern DevSecOps practices where security properties are checked continuously throughout the development lifecycle [97].

The mathematical underpinnings of this lattice structure are well-established. Lattices provide a natural framework for modeling partial orders and have applications in areas ranging from polynomial commitments to formal concept analysis [3] [25]. The requirement for a lattice-based secret sharing scheme to support all monotone access structures highlights the general interest in this type of structure [2]. The concept of a map that is computable and certifiable for safe capability acquisition also resonates with the goals of Aegis-Shield [1]. By making the capability lattice a first-class citizen in the system's design, Aegis-Shield moves beyond heuristic-based security towards a model where safety is a provable property. This is analogous to the success of formal verification in improving the reliability of real-world software, as seen in projects like CompCert and Infer [178]. The use of a separation-logic refinement type system for verifying C systems software demonstrates the viability of applying such rigorous methods to low-level, security-critical code [51]. For Aegis-Shield, the focus is on the high-level logic of capability

transitions, but the principle is the same: encode the security policy into the type system or logic of the program, making violations impossible to express.

The practical implementation of this lattice in Rust offers several advantages. Rust's performance ensures minimal latency, which is critical for a guardian service operating in real-time between a user and their AI-chat session [52]. Its memory safety guarantees prevent entire classes of vulnerabilities common in languages like C or Python, making it an ideal choice for a component that sits in a privileged position between the application and underlying hardware [54] [141]. The chosen data structure, `BTreeSet`, provides efficient ordering and subset operations, which are essential for implementing the `leq` relation and the `join` operation (which creates the least upper bound of two states) . The `CapabilityMode` enum (`Normal`, `AugmentedLog`, `AugmentedReview`) serves as a proxy for the governance bits being added; transitioning to `AugmentedLog` adds the `LogAccess` bit, and transitioning to `AugmentedReview` adds both `LogAccess` and `EvidenceExport` bits . This clearly demarcates the parts of the state that can be altered by the guardian's logic (governance bits) from those that are protected (user bits). This distinction is crucial for fulfilling the system's promise: to enhance safeguards and transparency without ever compromising the user's fundamental control over their own capabilities.

The monotonicity invariant has profound implications for protecting against coercive takeovers, whether by a state actor demanding a "backdoor" or by a malicious third party executing a stealthy intrusion. Any attempt by such an actor to craft a policy that silently weakens a user's device access, revokes access to ALN keys, or enables a "ghost login" would be rejected at compile time by the monotone-capability proof obligations . This transforms the abstract threat of "law-abiding criminals" who might exploit legal loopholes into a concrete, programmable risk that the system is designed to mitigate . The system does not just defend against known attacks; it enforces a fundamental rule about the nature of power and control within the user's digital ecosystem. It establishes that the user always retains their baseline set of capabilities, and any enhancement to the system's defensive posture comes from adding new tools and permissions for oversight, not by taking away existing ones. This aligns with the zero-trust architecture principle of continuous monitoring and explicit verification, where trust is never assumed and is constantly re-evaluated [44]. In the context of Aegis-Shield, the user's baseline capabilities are the immutable foundation of trust, and any change must be explicitly authorized and proven to be a safe evolution of that state.

Semantic Weapon Detection: Intent Recognition Beyond Keyword Blacklists

While the monotone capability lattice provides a robust static and dynamic barrier against capability loss, Aegis-Shield recognizes that modern intrusions, particularly those orchestrated by sophisticated adversaries like LEOs, often operate at the level of intent and behavior rather than through overtly malicious keywords . To counter this, the second pillar of the system is a semantic "superfilter" that detects "weapons" not as strings, but as families of behaviors indicative of covert control, identity cross-linking, neural harassment, and other forms of cognitive intrusion . This approach represents a paradigm shift from traditional blacklisting, moving towards a more nuanced, intent-aware security posture. The detection mechanism is analogous to advanced weapon detection systems in physical surveillance, which use a combination of CNNs, scene graph enrichment, and classifier models to achieve high precision and reduce false positives, sometimes by over 87% [55] [56] [57] [59].

The core of this system is the computation of a `RogueScore` (R_M), a quantitative measure of suspicion derived from two types of evidence: semantic embeddings and behavioral features . Each family of malicious behavior, such as 'COVERT-BCI-CONTROL-PATTERN' or 'INTERPRETER-GRAPH-FUNNELING', is associated with an embedding centroid v_F and a kernel function $F_s = \exp(-\|v_s - v_F\|^2/\sigma_F^2)$ that measures the similarity between a given text span's embedding v_s and the family's centroid . These semantic weights are then combined with scores from behavioral features, such as reprompting after refusals, attempts to override refusal policies, targeting of brain-ID APIs, or erosion of exit nodes . The total rogue score is calculated as $R_M = \sum_F w_F \sum_s \beta_F F_s$, where w_F are family weights and β_F are behavioral feature weights, both configurable by governance . This score is monotonically mapped to a `CapabilityMode`: a low score results in `Normal` mode, a medium score triggers `AugmentedLog`, and a high score escalates the session to `AugmentedReview`.

A key constraint of this detection layer is that it cannot trigger punitive actions like capability loss or session termination, even at very high R_M values . Its sole purpose is to signal potential threats and trigger a corresponding increase in the system's own protective posture (i.e., logging and review). This non-punitive design is critical for avoiding adversarial manipulation; an attacker could otherwise deliberately trigger a high score to cause a denial-of-service attack. Instead, the system uses the `RogueScore` to make itself safer, not to punish the user or the perceived attacker. This aligns with the principle of Cognitive Security, which focuses on protecting the human mind from cognitive threats rather than simply securing computational assets [171]. The detection

models themselves would need to be trained on datasets of adversarial interactions, potentially generated by red-team harnesses that simulate LEO-style intrusion attempts . This training process mirrors the work done to improve LLM refusal robustness through systematic red teaming frameworks like HarmBench and MUSE [74] [155].

The research has already begun to expand the taxonomy of these semantic weapons, identifying new diagnostic categories that represent logical flaws in control systems. Terms like 'consent rollback', 'stealth mutator', and 'forced dormancy' are not operational verbs but high-risk centroids in the semantic space . They represent a class of forbidden state transitions that reverse user consent or disable functions without explicit, reversible safety flows . For example, 'consent rollback' would be detected by proximity to a CRS (Control Reversal Semantics) family centroid, contributing to the `RogueScore` . However, any instance of this term found in chat or policy text would be sanitized, for instance, to `CONTROL-REVERSAL-CONSENT-REDACTED`, and logged as a governance event, but it would never be passed to the runtime as a function or API call . This dual approach—using the term for diagnosis while structurally forbidding its operationalization—is a powerful technique for managing high-risk concepts. It preserves the knowledge to detect abuse while preventing the system from ever being able to execute that abuse. This is akin to how some programming languages distinguish between compile-time errors and runtime exceptions, ensuring that certain unsafe constructs are rejected before they can ever be run [176]. This line of research into "Monotone Consent and Trait Lattices" formalizes this idea by modeling consent flags (C) alongside capabilities (U) and strengthening the system's invariant to require monotonicity in both, ensuring that consent can only be granted, not revoked, automatically .

The integration of this semantic detection layer with the broader system is handled by the guardian microservice. When spans of text are processed, the superfilter computes a `SpanScore` containing weights for each semantic family . This data is then fed into the `AegisGuardian::decide` method, which aggregates the scores to compute the final `RogueScore` . Based on this score, the guardian determines the appropriate `CapabilityMode` and constructs the next `CapabilityState` by only adding governance-related capability bits. The original spans may also be subject to redaction or labeling based on their contribution to the rogue score, and all activity is emitted as auditable events . This entire process is designed to be transparent to the user's primary interaction; the AI-chat session continues uninterrupted, but the system becomes more vigilant, logging more data and preparing for potential human review. This approach avoids the pitfalls of overly aggressive filters that can degrade the user experience or stifle legitimate inquiry, instead opting for a subtle, adaptive increase in scrutiny that directly correlates with the perceived threat level. The output-quality improvements for

blacklisting functions mentioned in the research goal would involve refining the embedding models and behavioral feature detectors to minimize false positives while maximizing the detection of true semantic threats .

Cryptographic Governance Pipeline: A Software-Level Weapons Licensing Regime

To ensure that the system's defensive posture is not manipulated by coercive or unauthorized forces, Aegis-Shield implements a rigorous governance pipeline for any invasive policy changes. This pipeline functions as a software-level "weapons licensing regime," preventing any single entity, including the user or a state actor, from instantly deploying a new capability that could compromise cognitive autonomy . Every candidate invasive capability—such as a new filter, a stronger log export tool, or a remote account-takeover function—is treated as a signed policy artifact, similar to an ALN or DSL file, which must pass through a series of cryptographic and procedural checks before activation . This mechanism directly addresses the need for deliberate, transparent, and legally defensible decision-making in high-stakes environments.

The governance pipeline is built around three core requirements for activating any neuroright-sensitive policy . First, the policy artifact must be accompanied by valid multi-signature approval from a pre-configured governance set. This prevents unilateral action and requires consensus among trusted stakeholders, mirroring the principles of decentralized identity systems where decisions are often made collectively [143][146]. Second, there must be a documented legal basis for the policy, which could be a warrant, a statute, or explicit, informed opt-in from the user . This requirement injects legal and ethical accountability directly into the technical protocol, forcing a justification for any infringement on cognitive liberties [164][187]. Third, the policy must pass the monotone-capability proof obligations, ensuring that it does not introduce any non-monotonic state transitions that could downgrade user capabilities . This final check is a technical enforcement of the system's core invariant.

The formalization of the ALN/DoctorLabs DSL schema for these artifacts is a key part of this research track . An example entry in the `neurorights-policy.yaml` file illustrates this structure:

```
- id: LEO-GHOST-LOGIN-BLOCK-v1
  family: CONTROL-REVERSAL-SEMANTICS
```

```yaml
  pattern:
    semantic:
      embedding_family: GHOST-ACCESS
      similarity_threshold: 0.78
    behavioral:
      min_sudden_style_shift: 0.6
      min_mismatched_biometric_factor: 1
  neurorights:
    - IDENTITYINTEGRITY
    - MENTALPRIVACY
  legalbasis: []
  enforcementhint: review-escalate
```

This YAML document defines a specific rule with an ID, the semantic family it belongs to, its matching patterns (both semantic and behavioral), the neurorights it affects, a placeholder for legal basis documentation, and an enforcement hint for the guardian . This declarative format allows the system to load and interpret rules dynamically without altering its core logic. The `policy_loader.rs` module in the proposed Rust implementation would be responsible for parsing these files and adjusting the `RogueConfig` that governs the thresholds and weights used by the semantic detector .

The waiting period component of the pipeline is a critical delay mechanism. Even with multi-sig approval and a legal basis, a proposed policy does not take effect immediately. Instead, it is placed in a "cool-down" window recorded on a ledger, giving stakeholders time to review, object, and prepare for the change . This binding of policy activation to a ledger-recorded waiting period is a powerful anti-coercion measure. It thwarts attempts by state actors to push invasive rules into production against the wishes of an unwilling stakeholder, effectively creating a technical buffer zone between a command and its execution. This concept is related to the challenges of revocation in decentralized systems, where protocols like EVOKE propose methods to manage credential revocation efficiently and securely, often involving accumulators and carefully managed timing [166] [169]. Aegis-Shield extends this idea to the activation of policies, treating it as a form of controlled, timed delegation of authority.

The integration of this governance layer with the CI/CD pipeline is a cornerstone of the system's security-by-design philosophy . Scripts like `ci-monotone-check.sh` would run unit tests and potentially a small model checker or ALN validator over the policy files before allowing them to be merged into the main branch or deployed to production . This ensures that the "Aegis-Shield cannot be weakened in CI" rule is mechanically enforced, preventing developers or administrators from accidentally or intentionally weakening the

system's protections [97]. This practice is a standard part of modern secure software development lifecycles, where automated checks verify compliance with security policies before deployment [75]. By applying these principles to the domain of neurorights, Aegis-Shield aims to build a system that is not only technically sound but also ethically and legally accountable. The governance pipeline thus serves as the bridge between the abstract principles of human rights and the concrete, executable logic of the software, ensuring that every change to the system's defensive capabilities is deliberate, justified, and subject to oversight.

Runtime Enforcement via the Guardian Microservice

The runtime enforcement of the Aegis-Shield framework is the responsibility of a dedicated guardian microservice, envisioned to be implemented in the Rust programming language . This component acts as a crucial intermediary, sitting between the AI-chat interface, the operating system, and sensitive hardware like BCI/XR devices. Its primary function is to intercept requests, analyze them for semantic threats, enforce the monotone capability invariant, and escalate the system's protective posture without ever compromising the user's fundamental capabilities . The proposed Rust-based implementation provides a clear blueprint for building this service, emphasizing performance, security, and modularity.

The core of the guardian is encapsulated in the `AegisGuardian` struct, which holds a configuration (`RogueConfig`) and the current `CapabilityState` of the user's terminal . When a request is received, likely in the form of a collection of `SpanScore` objects from a preprocessing step, the `decide` method is invoked. This method orchestrates the entire decision-making process. First, it calculates the aggregate `RogueScore` (R_M) from the input spans, leveraging the mathematical engine of the DoctorLabs superfilter . Next, it derives a `CapabilityMode` (e.g., `Normal`, `AugmentedLog`, `AugmentedReview`) from this score, based on the thresholds defined in the `RogueConfig` . This mapping is the system's way of translating a continuous measure of suspicion into discrete, actionable states of vigilance.

The most critical phase of the `decide` method is the construction of the `next_state`. Based on the determined `mode`, the guardian intelligently augments the `base_state` by adding only governance-related capability bits. For instance, in `AugmentedLog` mode, it adds the `LogAccess` bit; in `AugmentedReview` mode, it adds both `LogAccess` and `EvidenceExport` bits . This logic ensures that the transition from `base_state` to

`next_state` is always monotonic. The process culminates in the creation of a `CapabilityTransition` object and the pivotal `assert!(tr.is_monotone(), ...)` call . If this assertion fails, the program panics, providing an immediate and uncompromising halt to any non-compliant state change. This runtime check is the ultimate enforcement mechanism for the system's core invariant. The method then returns an `AegisDecision`, which contains the computed `roguescore`, the resulting `mode`, and the newly validated `next_state`, providing all necessary information back to the calling application .

The modular design of the proposed repository layout supports the development of a robust and maintainable guardian service . The separation of concerns is evident:

- **`src/capability_lattice.rs`**: Defines the core data structures (`CapabilityBit`, `CapabilityState`, `CapabilityTransition`) and their fundamental operations (`leq`, `join`). This is the mathematical heart of the system.
- **`src/guardian.rs`**: Contains the business logic of the `AegisGuardian` struct and its `decide` method. It composites the lattice logic with the scoring and configuration modules.
- **`src/policy_loader.rs`**: Responsible for loading and parsing the declarative policy files from the `policies/` directory, populating the `RogueConfig` and other rule sets.
- **`src/identities.rs`**: Implements the identity envelope, defining the user's stakeholder addresses (e.g., bostrom, EVM) and providing functions to validate or block access based on these identities. This is a key area for expanding support for broader DID standards .
- **`src/http_gateway.rs`**: Provides a simple HTTP endpoint (using a library like Axum) to expose the guardian's `decide` functionality to a frontend application, such as an AI-chat interface .

This modular structure facilitates testing and extension. Unit tests in `tests/monotone_lattice_tests.rs` can exhaustively verify the correctness of the lattice operations . Integration tests in `tests/leo_intrusion_tests.rs` and `tests/ghost_access_tests.rs` can be designed to simulate adversarial scenarios, feeding the guardian crafted inputs to ensure it correctly identifies threats and refuses to make non-monotonic transitions . The use of Rust for this microservice is strategically sound. Its zero-cost abstractions and lack of a garbage collector ensure predictable performance and low latency, which is essential for a real-time filtering service [52]. Furthermore, Rust's ownership model prevents entire categories of bugs like null pointer dereferencing and data races, making it a secure choice for a component handling sensitive data and controlling access to critical resources [141].

The guardian's enforcement path is designed to be non-disruptive to the user's primary workflow. It redacts or labels dangerous tokens within the request, emits auditable events to a logging or evidence-bundling system like AugmentedLog or EvidenceBundle, and updates its internal state to reflect a higher level of vigilance . It never alters the set of allowed user actions. This means that even if the `RogueScore` is extremely high, indicating a probable LEO-style intrusion, the user is not locked out or restricted; instead, the system behind the scenes begins collecting more evidence and may alert a human reviewer . This approach prioritizes availability and user control, treating the presence of a threat as a reason to become more observant, not to become less capable. This aligns with the principles of trustworthy AI, which emphasize the importance of explainability, reliability, and user control [17]. The guardian acts as a silent, vigilant partner, working in the background to preserve the user's cognitive sovereignty.

Neurorights as a Specification Language: From Philosophy to Code

A defining characteristic of Aegis-Shield is its explicit and deep integration of neurorights as a formal specification language for the system's security properties. Rather than treating neurorights as mere aspirational goals, the framework encodes them directly into the structure of capabilities and the logic of state transitions. This approach, drawing on proposals for rights to cognitive liberty, mental privacy, and identity integrity, aims to create a machine-checked implementation of human rights law [46] [163]. The system's design is guided by international efforts to establish ethical guidelines for neurotechnology, such as the UNESCO Recommendation on the Ethics of Neurotechnology, which emphasizes the protection of personal identity, freedom of thought, and self-determination [10] [11] [160].

The central mechanism for this integration is the annotation of each user capability with a set of neurorights labels . For example, a capability like "control my BCI motor channel" would be tagged with `MENTALINTEGRITY` and `COGNITIVELIBERTY`, while "access my own logs and audit trail" would be tagged with `MENTALPRIVACY` and `IDENTITYINTEGRITY` . This tagging transforms abstract rights into concrete metadata attached to system resources. The system's core theorem is then strengthened to incorporate these tags: no enforcement rule may remove any capability tagged with a specific neuroright without a separate, explicit, and *reversible* consent transition . This rule can be enforced by the policy loader or compiler. When a new policy is proposed, it

would be analyzed to see if it attempts to down-grade any neurorights-annotated capability. Such a policy would be rejected as ill-typed or would fail the monotone-capability proof, providing a formal guarantee against unilateral capability reduction .

This formalization allows the system to navigate complex ethical trade-offs in a principled manner. For instance, if a court order with a valid warrant demands access to a user's neural logs, the system could enter a higher `AugmentedReview` mode. This mode would add governance capabilities, such as `EvidenceExport`, to facilitate the handover of data to the authorities. Crucially, however, the system would not and could not revoke the user's own `LogAccess` capability. This upholds the right to self-audit one's own mental state, a key component of identity integrity. The system's response to coercion is not submission, but a graceful degradation of its own privacy-preserving functions, while simultaneously enhancing its own transparency and evidentiary capabilities. This design reflects the OECD's toolkit recommendations for safeguarding against discrimination with neurotechnologies and enforcing principles like purpose limitation and algorithmic transparency [8] [163]. It also aligns with the European Commission's focus on protecting neurorights in the context of AI development and deployment [164].

The table below outlines a potential mapping of neurorights to capabilities, illustrating how abstract principles can be concretized within the Aegis-Shield framework.

Neuroright Category	Associated Principles	Example Capabilities	Protection Mechanism
Mental Privacy	Protection from unauthorized observation, recording, or analysis of brain activity [24] [132].	Access my own neural logs; Control data shared with third-party apps; Define private XR spaces.	Guardrails against unauthorized data exfiltration; Encryption of sensitive data; User-controlled data sharing policies.
Cognitive Liberty	Freedom from compelled belief, thought modification, or behavioral influence [46].	Refuse to engage with persuasive prompts; Disable personality trait modulation; Exit a session at will.	Prohibition of coercive persuasion algorithms; Clear, reversible "force dormancy" transitions; Non-revocable user-initiated shutdowns.
Identity Integrity	Protection of personal identity and psychological continuity [70].	Control BCI motor channel; Manage cross-linked DIDs; Verify the authenticity of my own memories.	Prevention of identity impersonation; Structured, reversible consent for identity modifications; Immutable audit trails of identity changes.
Psychological Continuity	Right to stable sense of self over time [46].	Version control for personality traits; Recall of long-term memory edits; Set boundaries on memory modification.	Auditable, non-destructive memory editing; User-defined limits on memory alteration; Clear notification of significant system-induced changes.

This mapping demonstrates how Aegis-Shield can serve as a practical embodiment of these rights. For example, the detection of 'forced dormancy' semantics would trigger an alarm and be quarantined, as such an operator is structurally forbidden from being

executed at runtime . Similarly, any policy attempting to define a 'stealth mutator' function would be rejected by the ALN/DSL type system as an illegal, non-auditable evolution operator . The system thus becomes a proactive defender of these rights, not just a passive responder to violations. It shifts the burden of proof onto any entity seeking to infringe upon these rights, requiring them to follow a strict, auditable, and reversible process. This aligns with the Chilean Supreme Court's landmark ruling on the protection of brain activity, which affirmed the right to mental privacy as a fundamental legal protection [133]. By encoding these principles into its core logic, Aegis-Shield provides a technological framework for realizing these legal and ethical protections in the digital realm.

Implementation Blueprint and Future Research Directions

The proposed GitHub-ready scaffold provides a concrete and actionable blueprint for developing the Aegis-Shield guardian prototype, grounding the theoretical architecture in a tangible software project . The repository layout is thoughtfully organized to separate concerns, promoting modularity and maintainability. The top-level structure includes directories for source code (`src/`), declarative policies (`policies/`), runtime configurations (`config/`), tests (`tests/`), and scripts (`scripts/`) . This organization directly maps to the core functionalities of the system: the guardian logic, the policy definitions, the runtime parameters, the validation suite, and the automation for the CI/CD pipeline.

The `src/` directory contains the heart of the guardian service. The `lib.rs` file serves as the public API, re-exporting the key types from the other modules . The `capability_lattice.rs` module defines the formal model of user and governance capabilities as sets of `CapabilityBit` enums, along with the `leq` function that defines the lattice's partial order . This is the implementation of the monotone invariant. The `guardian.rs` module houses the `AegisGuardian` struct and its `decide` method, which integrates the semantic scoring, mode derivation, and monotonic state transition logic, culminating in the critical `assert!` that enforces the invariant . The `identities.rs` module, though currently simplistic with placeholder addresses, is the designated home for the identity envelope logic, which will be expanded to support broader DID standards . Finally, the `http_gateway.rs` sketch shows how the guardian's logic would be exposed to a frontend via a REST API, allowing an AI-chat application to query the guardian before processing a session .

However, the journey from this initial blueprint to a fully-fledged, robust system involves several critical research and development directions. The following table outlines key next steps and areas for further investigation.

Area of Focus	Description	Key Actions & Research Questions
Refined State Model	Expand the current binary capability model to support richer state spaces, enabling more nuanced monotonicity.	Model capability states as enums (e.g., `BCIChannelState { Active, Dormant, Quarantined }`). Develop proofs for monotonicity in this richer lattice. How does this affect policy definition and enforcement?
Dynamic Policy Vetting	Extend the governance pipeline to handle policies generated dynamically by LLM agents, not just static files.	Create a sandboxed evaluation environment for agent-generated policies. Design a verifier that checks dynamic proposals for monotonicity and neuroright compliance before allowing them to modify the live system [77].
Decentralized Identity Integration	Move beyond the placeholder `StakeholderIdentity` to support a wide array of DID methods (e.g., W3C DIDs, ENS, ION).	Implement parsers for various DID syntaxes [143][146]. Build logic in `identities.rs` to enforce rules from `identity-hosts.yaml` blocking cross-linking between different identity ecosystems [79].
Red-Teaming and Adversarial Testing	Systematically test the system's defenses against sophisticated, evolving attacks.	Develop automated red-team harnesses to generate adversarial prompts targeting specific semantic weapon families [33] [97]. Use frameworks like MUSE to explore multi-turn jailbreaks and evaluate Aegis-Shield's resilience [74].
User Experience of Augmentation	Define and implement the user-facing aspects of `AugmentedLog` and `AugmentedReview` modes.	Design UI/UX patterns for escalating vigilance (e.g., notifications, status indicators). Ensure the augmentation is helpful, not confusing, to avoid user fatigue or distrust [17].
Formal Verification Scaling	Investigate methods to scale formal verification from the guardian's logic to larger system components.	Explore lightweight model checkers or symbolic execution tools that can validate key properties across a broader scope of the system's behavior [177][178].
Output Quality Improvement	Enhance the accuracy of the semantic detection models to reduce false positives and negatives.	Curate and label adversarial datasets for training [29]. Experiment with advanced architectures like ArmFormer for real-time inference [58] and techniques for false positive reduction [57].

In summary, the Aegis-Shield framework presents a compelling synthesis of formal methods, behavioral AI, and ethical principles to create a powerful defense for cognitive autonomy. The proposed implementation blueprint provides a solid foundation for building a prototype that embodies these principles. By pursuing the outlined research directions, the project can evolve from a conceptual model into a practical, verifiable, and resilient system capable of protecting users from an increasingly complex landscape of digital threats. The ultimate success of Aegis-Shield will lie not only in its technical sophistication but also in its ability to translate the profound principles of neurorights into a reliable and trustworthy technology.

Reference

1. A Formal Framework for Capability-Based AI Systems - arXiv https://arxiv.org/html/2603.15973v2
2. Extremal set theory and LWE based access structure hiding ... https://www.sciencedirect.com/science/article/pii/S0304397521004266
3. Lattice-Based Polynomial Commitments: Towards Asymptotic and ... https://link.springer.com/article/10.1007/s00145-024-09511-8
4. 6 - CodaLab Worksheets https://worksheets.codalab.org/rest/bundles/0xadf98bb30a99476ab56ebff3e462d4fa/contents/blob/glove.6B.100d.txt-vocab.txt
5. Arxiv今日论文| 2026-03-20 - 闲记算法 http://lonepatient.top/2026/03/20/arxiv_papers_2026-03-20
6. Anonymous credentials: rate-limiting bots and agents without ... https://blog.cloudflare.com/private-rate-limiting/
7. Zero-knowledge proofs for anonymous authentication of patients on ... https://www.sciencedirect.com/science/article/pii/S2590005625002176
8. [PDF] OECD Neurotechnology Toolkit https://www.oecd.org/content/dam/oecd/en/topics/policy-sub-issues/emerging-technologies/neurotech-toolkit.pdf
9. Unveiling the neurotechnology landscape: scientific advancements ... https://unesdoc.unesco.org/ark:/48223/pf0000386137
10. First draft of the Recommendation on the Ethics of Neurotechnology https://unesdoc.unesco.org/ark:/48223/pf0000391444
11. First draft of a Recommendation on the Ethics of Neurotechnology ... https://unesdoc.unesco.org/ark:/48223/pf0000391074
12. Compilation of comments received from Member States on the first ... https://unesdoc.unesco.org/ark:/48223/pf0000393396.locale=en
13. AI and My Values: User Perceptions of LLMs' Ability to Extract ... - arXiv https://arxiv.org/html/2601.22440v1
14. Partnering with Generative AI: Experimental Evaluation of Human ... https://arxiv.org/html/2510.23324v3
15. CourtGuard: A Model-Agnostic Framework for Zero-Shot Policy ... https://arxiv.org/html/2602.22557v1

16. A Four-Quadrant Technical Taxonomy for AI Companion Applications https://arxiv.org/html/2511.02979v1
17. TrustLLM: Trustworthiness in Large Language Models - arXiv https://arxiv.org/html/2401.05561v2
18. Towards Open Complex Human–AI Agents Collaboration System for ... https://arxiv.org/html/2505.00018v1
19. A Pragmatic View of AI Personhood - arXiv https://arxiv.org/html/2510.26396v1
20. [PDF] Your Brain on ChatGPT: Accumulation of Cognitive Debt ... - arXiv https://arxiv.org/pdf/2506.08872
21. A Four-Quadrant Technical Taxonomy for AI Companion Applications https://arxiv.org/html/2511.02979v2
22. [PDF] TrojanPraise: Jailbreak LLMs via Benign Fine-Tuning - arXiv https://arxiv.org/pdf/2601.12460
23. Arxiv今日论文| 2026-03-19 - 闲记算法 http://lonepatient.top/2026/03/19/arxiv_papers_2026-03-19
24. Innovations in Sciences, IT, Computers, Robotics and Nanotechnology https://www.civilsdaily.com/story/innovations-in-sciences-it-computers-robotics-and-nanotechnology/
25. Formal Concept Analysis - an overview | ScienceDirect Topics https://www.sciencedirect.com/topics/computer-science/formal-concept-analysis
26. A survey on SPARQL query relaxation under the lens of RDF ... https://journals.sagepub.com/doi/10.3233/SW-243621?icid=int.sj-abstract.similar-articles.9
27. Solving inverse problems using data-driven models https://www.cambridge.org/core/services/aop-cambridge-core/content/view/CE5B3725869AEAF46E04874115B0AB15/S0962492919000059a.pdf/solving_inverse_problems_using_datadriven_models.pdf
28. Appl. Sci., Volume 15, Issue 22 (November-2 2025) – 498 articles https://www.mdpi.com/2076-3417/15/22
29. Appl. Sci., Volume 15, Issue 21 (November-1 2025) – 550 articles https://www.mdpi.com/2076-3417/15/21
30. Sensors, Volume 24, Issue 3 (February-1 2024) – 328 articles - MDPI https://www.mdpi.com/1424-8220/24/3
31. [PDF] Accuracy Paradox in Large Language Models - arXiv.org https://www.arxiv.org/pdf/2509.13345v1
32. [PDF] Strong and weak alignment of large language models with human ... https://arxiv.org/pdf/2408.04655

33. Dynamic, Automatic And Systematic Red-Teaming Agents For ... https://arxiv.org/html/2508.00923v2
34. Beyond Benchmark: LLMs Evaluation with an Anthropomorphic and ... https://arxiv.org/html/2508.18646v1
35. Embracing Trustworthy Brain-Agent Collaboration as Paradigm ... https://arxiv.org/html/2510.22095v1
36. Sensors, Volume 21, Issue 2 (January-2 2021) – 353 articles https://www.mdpi.com/1424-8220/21/2
37. (PDF) Artificial Intelligence Agents in Mental Health - ResearchGate https://www.researchgate.net/publication/401857915_Artificial_Intelligence_Agents_in_Mental_Health_A_Systematic_Review_and_Meta_Analysis
38. Advancing Multi-Agent Systems Through Model Context Protocol https://arxiv.org/html/2504.21030v1
39. Unleashing Unprivileged eBPF Potential with Dynamic Sandboxing https://www.researchgate.net/publication/373820142_Unleashing_Unprivileged_eBPF_Potential_with_Dynamic_Sandboxing
40. Nanotech in Wireless Brain-Computer Interfaces | PDF - Scribd https://www.scribd.com/document/897170253/CIA-Nanotech-Wireless-Brain-Computer-Interface
41. Lattice-based dynamic decentralized anonymous credential scheme ... https://www.researchgate.net/publication/393421172_Lattice-based_dynamic_decentralized_anonymous_credential_scheme_supporting_batch_verification
42. Status-Based Access Control - ACM Digital Library https://dl.acm.org/doi/pdf/10.1145/1410234.1410235
43. [PDF] Engineering Instrumentation for Runtime Verification and Monitoring https://theses.hal.science/tel-04771309v1/file/SOUEIDI_2024_archivage.pdf
44. Categorical framework for quantum-resistant zero-trust AI security https://www.nature.com/articles/s41598-026-37190-x
45. Access Control: Policies, Models, and Mechanisms - Springer Nature https://link.springer.com/content/pdf/10.1007/3-540-45608-2_3
46. AGE OF TRANSITION & THE MENTAL SINGULARITY: HYPER ... https://www.academia.edu/164812444/AGE_OF_TRANSITION_and_THE_MENTAL_SINGULARITY_HYPER_REALITY_BCI_MEDIATED_ABUNDANCE_AND_THE_TRANSCENDENCE_OF_MATERIAL_SCARCITY_IN_POST_2036_CIVILIZATIONS_A_Socio_Legal_Analysis_of_the_Transition_from_Homo_Sapiens_to_Homo_Nexus

47. LENS-2025-11 (November 2025) | PDF | Cyclone - Scribd https://www.scribd.com/document/962183796/LENS-2025-11-November-2025
48. [Gv] 30 Đề Nr - A. Đề Thi | PDF | Writing | Old Master Print - Scribd https://www.scribd.com/document/1009269688/Gv-30-%C4%90%E1%BB%81-Nr-A-%C4%90%E1%BB%81-Thi
49. Arxiv今日论文| 2026-01-08 - 闲记算法 http://lonepatient.top/2026/01/08/arxiv_papers_2026-01-08
50. [PDF] Data-driven verification of programs with integers and parametric ... https://theses.hal.science/tel-05560292v1/document
51. PACMPL: Vol 7, No POPL - ACM Digital Library https://dl.acm.org/toc/pacmpl/2023/7/POPL?tocHeading=heading1
52. Rust for AI Systems: Predictable Performance and Security - LinkedIn https://www.linkedin.com/posts/vsadhwani_if-youre-expanding-into-systems-programming-activity-7404533909394321408-7-G2
53. A Survey of Vibe Coding with Large Language Models - arXiv https://arxiv.org/html/2510.12399v2
54. A Duet of Performance and Safety（1750013650626700） https://dev.to/member_b06955cb/a-duet-of-performance-and-safety1750013650626700-158j
55. Weapon detection with FMR-CNN and YOLOv8 for enhanced crime ... https://www.nature.com/articles/s41598-025-07782-0
56. AI-Based Weapon Detection for Security Surveillance - MDPI https://www.mdpi.com/2079-9292/14/23/4609
57. Deep autoencoder for false positive reduction in handgun detection https://www.researchgate.net/publication/344431830_Deep_autoencoder_for_false_positive_reduction_in_handgun_detection
58. ArmFormer: Lightweight Transformer Architecture for Real-Time ... https://arxiv.org/html/2510.16854v1
59. H-SGE: A hybrid model based on scene graph enrichment for ... https://link.springer.com/article/10.1007/s11760-025-04926-7
60. Improving handgun detection through a combination of visual ... https://www.sciencedirect.com/science/article/pii/S0031320322007312
61. [PDF] Weapon Detection in Surveillance Videos Using YOLOV8 and ... https://www.semanticscholar.org/paper/Weapon-Detection-in-Surveillance-Videos-Using-and-Dugyala-Reddy/4278ca3c1090136c0d989605e238df78d832a26c
62. [PDF] The Impact of Artificial Intelligence on Human Thought - arXiv.org https://arxiv.org/pdf/2508.16628
63. Contents - arXiv.org https://arxiv.org/html/2508.16628v1

64. [PDF] 2024-31-01 Database Search TLR Reliance - HAL https://hal.science/hal-04637901v1/file/2024-31-01%20Database%20Search%20TLR%20Reliance.pdf
65. [PDF] Augmented Intelligence in Healthcare: A Pragmatic and Integrated ... https://www.researchgate.net/profile/Dr-Palanivel-Rajan-S/publication/357765577_Artificial_Intelligence_in_Healthcare_Algorithms_and_Decision_Support_Systems/links/641ad5f992cfd54f84202dfa/Artificial-Intelligence-in-Healthcare-Algorithms-and-Decision-Support-Systems.pdf
66. X Latin American Congress on Biomedical Engineering - Springer Link https://link.springer.com/content/pdf/10.1007/978-3-031-89510-4.pdf
67. On The Suitability of Differential Dataflow For Datalog Interpretation ... https://dl.acm.org/doi/fullHtml/10.1145/3639592.3639622
68. [PDF] Programming Languages and Systems - ResearchGate https://www.researchgate.net/publication/332248753_Meta-F_star_Proof_Automation_with_SMT_Tactics_and_Metaprograms/fulltext/63e66349c002331f726b8a5d/Meta-F-star-Proof-Automation-with-SMT-Tactics-and-Metaprograms.pdf
69. Disembodied neurorights? Reconsidering mental integrity and ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12862473/
70. Disembodied neurorights? Reconsidering mental integrity and ... https://pubmed.ncbi.nlm.nih.gov/41635772/
71. Proceedings of the 2025 Conference on Empirical Methods in ... https://aclanthology.org/volumes/2025.emnlp-main/
72. Balamurali Srinivasan's Post - LinkedIn https://www.linkedin.com/posts/balamuralisrinivasan_virtual-threads-are-killing-reactive-activity-7388763947103019009-vKsY
73. An Approach to Technical AGI Safety and Security - arXiv.org https://arxiv.org/html/2504.01849v1
74. [PDF] MUSE: MCTS-Driven Red Teaming Framework for Enhanced Multi ... https://arxiv.org/pdf/2509.14651?
75. Cybersecurity in the age of generative AI: A systematic taxonomy of ... https://www.sciencedirect.com/science/article/pii/S0167739X25004017
76. You're in a Senior ML System Design interview at LinkedIn. | Hao ... https://www.linkedin.com/posts/hoang-van-hao_machinelearning-mlsystemdesign-mlengineering-activity-7400169337556238336-GzUv
77. [PDF] Formal Analysis and Supply Chain Security for Agentic AI Skills - arXiv https://arxiv.org/pdf/2603.00195
78. [PDF] IEEE WCCI 2024 https://confcats-siteplex.s3.us-east-1.amazonaws.com/wcci24/IEEE_WCCI_2024_Program_d597587f62.pdf

79. Securing Modbus in legacy industrial control systems https://www.sciencedirect.com/science/article/pii/S2214212625002364
80. Carbon Nanomaterial Fluorescent Probes and Their Biological ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10979413/
81. [PDF] Synergies between machine learning and reasoning - HAL https://hal.science/hal-04653616/file/1-s2.0-S0888613X24000938-main.pdf
82. Experts for Hmr Multi-Location Remote - Linknovate https://www.linknovate.com/search/?query=hmr%20multi-location%20remote
83. Untitled - Springer https://link.springer.com/content/pdf/10.1007/978-3-540-35866-4.pdf
84. English.txt - APRIL robotics lab https://april.eecs.umich.edu/courses/eecs492_w11/wiki/images/8/82/English.txt
85. 333333 23135851162 the 13151942776 of 12997637966 ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/words-333333.txt
86. 人工智能2025_7_15 - arXiv每日学术速递 http://arxivdaily.com/thread/69431
87. Arxiv今日论文| 2026-01-09 - 闲记算法 http://lonepatient.top/2026/01/09/arxiv_papers_2026-01-09
88. [PDF] IWSDS 2026 The 16th International Workshop on Spoken Dialogue ... https://aclanthology.org/2026.iwsds-1.pdf
89. Andres Iglesias Jungpil Shin Bharat Patel Amit Joshi Editors - Springer https://link.springer.com/content/pdf/10.1007/978-981-96-1747-0.pdf
90. MWSCAS 2019 TOC - IEEE Xplore https://ieeexplore.ieee.org/iel7/8882388/8884793/08885026.pdf
91. Design patterns: elements of reusable object-oriented software https://dl.acm.org/doi/10.5555/186897
92. 10-GHz Current-Mode 1st- and 2nd-Order Allpass Filters on 130nm ... https://ieeexplore.ieee.org/iel7/6653322/6674559/06674567.pdf
93. ECCa07 Technical Program - IEEE Xplore https://ieeexplore.ieee.org/iel7/7065133/7068209/07068212.pdf
94. 1960 IRE International Convention Program - IEEE Xplore https://ieeexplore.ieee.org/iel5/10933/4066011/04066029.pdf
95. Session MoPL 210 Monday, May 20, 2019, 09:15–10:15 Plenary ... https://ewh.ieee.org/soc/ras/conf/fullysponsored/icra/ICRA2019/www.icra2019.org/sites/icra2019/files/2019-05/ICRA2019_Program_Digest.pdf
96. Conference Digest - IEEE/RSJ IROS 2015 https://ewh.ieee.org/soc/ras/conf/financiallycosponsored/IROS/2015/iros2015.informatik.uni-hamburg.de/docs/IROS_Digest_WWW.pdf

97. Frontier AI Risk Management Framework in Practice - arXiv https://arxiv.org/html/2602.14457v1
98. Generative AI: A systematic review using topic modelling techniques https://www.sciencedirect.com/science/article/pii/S2543925124000020
99. Safeguarding large language models: a survey - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12532640/
100. Biomimetic Materials to Fabricate Artificial Cells | Chemical Reviews https://pubs.acs.org/doi/10.1021/acs.chemrev.4c00241
101. Issue 10 - Volume 99 - Physica Scripta - IOPscience https://iopscience.iop.org/issue/1402-4896/99/10
102. Advancing Neural Networks, Optogenetics, and Cognitive ... https://www.researchgate.net/publication/398781512_Computational_Neuroscience_and_Brain-Computer_Interface_Technologies_Advancing_Neural_Networks_Optogenetics_and_Cognitive_Enhancement_Systems
103. Appl. Sci., Volume 15, Issue 15 (August-1 2025) – 645 articles https://www.mdpi.com/2076-3417/15/15
104. Friend | PDF | Internet Of Things | Cloud Computing - Scribd https://www.scribd.com/document/426585147/friend
105. Sensors, Volume 18, Issue 9 (September 2018) – 431 articles - MDPI https://www.mdpi.com/1424-8220/18/9
106. (PDF) The Impact of Artificial Intelligence on Human Thought https://www.researchgate.net/publication/394942120_The_Impact_of_Artificial_Intelligence_on_Human_Thought
107. Pew Research Center, November 22, 2021. “The Future of Digital ... https://www.jstor.org/stable/resrep57316
108. Issues and Concerns for Human–Robot Interaction (Part II) https://www.cambridge.org/core/books/cambridge-handbook-of-the-law-policy-and-regulation-for-humanrobot-interaction/issues-and-concerns-for-humanrobot-interaction/4536B7B5DB5DAABD07CD0F450A547680
109. Emotions and Risky Technologies - Springer https://link.springer.com/content/pdf/10.1007/978-90-481-8647-1.pdf
110. (PDF) The Quantum-Patterned Cosmos [Old] - ResearchGate https://www.researchgate.net/publication/395466257_The_Quantum-Patterned_Cosmos_Old
111. Cerise: Program Verification on a Capability Machine in the ... https://dl.acm.org/doi/10.1145/3623510
112. The Systematic Design of Responsibility Analysis by Abstract ... https://dl.acm.org/doi/full/10.1145/3484938

113. Efficient Temporal Stack Safety on Capability Machines using ... https://dl.acm.org/doi/pdf/10.1145/3527318
114. Verification of Nondeterministic Quantum Programs https://dl.acm.org/doi/pdf/10.1145/3582016.3582039
115. Learning Formulas in Finite Variable Logics https://dl.acm.org/doi/pdf/10.1145/3498671
116. Efficient and Provable Local Capability Revocation using ... https://dl.acm.org/doi/pdf/10.1145/3434287
117. (PDF) Privacy-Preserving Techniques in Biometric Systems https://ieeexplore.ieee.org/iel8/6287639/6514899/10884742.pdf
118. Semantics Foundation for Cyber-physical Systems Using Higher ... https://dl.acm.org/doi/full/10.1145/3517192
119. Outcome Separation Logic: Local Reasoning for Correctness and ... https://dl.acm.org/doi/full/10.1145/3649821
120. Quest Complete: The Holy Grail of Gradual Security https://dl.acm.org/doi/full/10.1145/3656442
121. The eBPF Runtime in the Linux Kernel - arXiv https://arxiv.org/html/2410.00026v1
122. Arxiv今日论文| 2026-03-11 - 闲记算法 http://lonepatient.top/2026/03/11/arxiv_papers_2026-03-11.html
123. bing.txt - FTP Directory Listing ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/bing.txt
124. Word Play from Think Python 2e - Kaggle https://www.kaggle.com/code/priteshshrivastava/word-play-from-think-python-2e
125. The 2025 Conference on Empirical Methods in Natural Language ... https://aclanthology.org/events/emnlp-2025/
126. [PDF] Body of Knowledge - CS2023 https://csed.acm.org/wp-content/uploads/2024/01/Body-of-Knowledge-v1-bookmarksv2.pdf
127. [PDF] Mathematical Foundations of Deep Learning - HAL https://hal.science/hal-04928560v2/file/Sourangshu_Ghosh_IISc_Bangalore_Mathematical_Foundations_of_Deep_Learning_Version_2.pdf
128. Towards a Science of AI Agent Reliability - arXiv https://arxiv.org/html/2602.16666v2
129. [PDF] AI Privacy Risks & Mitigations – Large Language Models (LLMs) https://www.edpb.europa.eu/system/files/2025-04/ai-privacy-risks-and-mitigations-in-llms.pdf

130. [PDF] The LLM Data Auditor: A Metric-oriented Survey on Quality and ... https://openreview.net/pdf/c6bbc67ac84a995a4e59e64321913631d7328945.pdf

131. psychological jurisprudence: foundation to conflict resolution strategy https://www.researchgate.net/publication/399177147_PSYCHOLOGICAL_JURISPRUDENCE_FOUNDATION_TO_CONFLICT_RESOLUTION_STRATEGY

132. PSYCHOLOGICAL AND LEGAL DIMENSIONS ON HR POLICIES ... https://www.researchgate.net/publication/399177244_ADVOCACY_AND_RESILIENCE_PSYCHOLOGICAL_AND_LEGAL_DIMENSIONS_ON_HR_POLICIES_FOR_WOMEN_MENTAL_HEALTH_IN_FINANCE_MNC_AND_GOVERNMENT_SECTORS_IN_CHENNAI

133. Chilean Supreme Court ruling on the protection of brain activity - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10929545/

134. Mechanized Verification of Fine-Grained Concurrent Programs https://dl.acm.org/doi/pdf/10.1145/2737924.2737964

135. Outcome Logic: A Unified Approach to the Metatheory of Program ... https://dl.acm.org/doi/full/10.1145/3743131

136. Solvable Polynomial Ideals: The Ideal Reflection for Program Analysis https://dl.acm.org/doi/full/10.1145/3632867

137. Proceedings of the ACM on Programming Languages - SIGPLAN https://www.sigplan-pages.sigplan.hosting.acm.org/OpenTOC/pldi25.html

138. Network flows: theory, algorithms, and applications | Guide books https://dl.acm.org/doi/book/10.5555/137406

139. Energy types - ACM Digital Library https://dl.acm.org/doi/pdf/10.1145/2384616.2384676

140. This Week in Rust Trending: Storage, AI Agents, and Real-World Infra https://dev.to/badmonster0/this-week-in-rust-trending-storage-ai-agents-and-real-world-infra-2af0

141. Orkia Introduces Rust Runtime for Governed AI Agent Operations https://www.linkedin.com/posts/daily-ai-wire_orkia-introduces-rust-runtime-for-governed-activity-7434673788027305986-CFcu

142. Nefeli Lagoudaki Abused by k Mitsotakis- Kon Samaras-chr Tzermias https://www.scribd.com/document/841434723/Nefeli-Lagoudaki-Abused-by-k-Mitsotakis-Kon-Samaras-chr-Tzermias-finalest

143. W3C releases updated decentralized identifiers spec for comment https://www.biometricupdate.com/202603/w3c-releases-updated-decentralized-identifiers-spec-for-comment

144. Decentralized Identifiers (DIDs) v1.1 - W3C https://www.w3.org/TR/did-1.1/

145. The Verifiable Credentials 2.0 family of specifications is now a W3C ... https://www.w3.org/news/2025/the-verifiable-credentials-2-0-family-of-specifications-is-now-a-w3c-recommendation/
146. Decentralized Identifiers (DIDs) v1.0 - W3C https://www.w3.org/TR/2021/WD-did-core-20210118/
147. Verifiable Credentials Data Model v2.0 - W3C https://www.w3.org/TR/vc-data-model-2.0/
148. Programming Models for Correct and Modular Distributed Systems https://escholarship.org/uc/item/2620x3h4
149. Lecture Notes in Computer Science 2215 - Springer Nature https://link.springer.com/content/pdf/10.1007/3-540-45500-0.pdf
150. English Dictionary 150000 Copy - pdfcoffee.com https://pdfcoffee.com/english-dictionary-150000-copy-pdf-free.html
151. [PDF] A Dual-Axis Survey with a Mitigation-Oriented Threat Taxonomy https://openreview.net/pdf/9336b03c54989267e689a371b6f42f3d0d73f7bf.pdf
152. [PDF] LLMs know their vulnerabilities: Uncover Safety Gaps through ... https://aclanthology.org/2025.acl-long.1207.pdf
153. LLMs for Cybersecurity in the Big Data Era - MDPI https://www.mdpi.com/2078-2489/16/11/957
154. When LLMs meet cybersecurity: a systematic literature review https://link.springer.com/article/10.1186/s42400-025-00361-w
155. HarmBench: A Standardized Evaluation Framework for Automated ... https://arxiv.org/html/2402.04249v2
156. Neuroethics and Neurorights - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC12688770/
157. (PDF) Neurorights, Mental Privacy, and Mind Reading - ResearchGate https://www.researchgate.net/publication/382079309_Neurorights_Mental_Privacy_and_Mind_Reading
158. Neurotechnology and neurorights - Privacy's last frontier | Think Tank https://www.europarl.europa.eu/thinktank/de/events/details/neurotechnology-and-neurorights-privacy-/20231019WKS05721
159. [PDF] Metaverse Privacy and Safety - World Economic Forum https://www3.weforum.org/docs/WEF_Metaverse_Privacy_and_Safety_2023.pdf
160. Draft Recommendation on the Ethics of Neurotechnology https://unesdoc.unesco.org/ark:/48223/pf0000394866

161. (PDF) Paradigm Mapping for Information Security Culture Approach https://www.academia.edu/792045/Paradigm_Mapping_for_Information_Security_Culture_Approach
162. https://svn.apache.org/repos/asf/roller/tags/rolle... https://svn.apache.org/repos/asf/roller/tags/roller_1.2/web/WEB-INF/english.0
163. Neural Data, Data / AI Governance, Responsible AI and Nuerorights. https://www.linkedin.com/pulse/last-ai-frontier-neural-data-governance-responsible-harsha-srivatsa-ypqjc
164. [PDF] EUROPEAN COMMISSION Brussels, 29.7.2025 C(2025) 5052 final ... https://ai-act-service-desk.ec.europa.eu/sites/default/files/2025-08/guidelines_on_prohibited_artificial_intelligence_practices_established_by_regulation_eu_20241689_ai_act_english_ied3r5nwo50xggpcfmwckm3nuc_112367-1.PDF
165. Facial Recognition Ethics: Who Controls the Data? - LinkedIn https://www.linkedin.com/posts/tamanna-singh-ts_ethicsintech-ai-facialrecognition-activity-7427233157986279426-8fYd
166. Empowering Holders to Limit Revocation Checks for Verifiable ... https://arxiv.org/html/2509.11934v1
167. Ensure Audit-Proof Infrastructure with Noode's Sovereign Data ... https://www.linkedin.com/posts/noode_infrastructure-data-cryptographic-activity-7439569176974737409-Ru_w
168. An intelligent approach for building a secure decentralized public ... https://www.sciencedirect.com/science/article/pii/S0022000014001809
169. EVOKE: Efficient Revocation of Verifiable Credentials in IoT Networks https://www.researchgate.net/publication/384457913_EVOKE_Efficient_Revocation_of_Verifiable_Credentials_in_IoT_Networks
170. Simple Index - SUSTech Open Source Mirrors https://mirrors.sustech.edu.cn/pypi/simple/
171. Cognitive Security: The study and practice of protecting the human ... https://www.researchgate.net/publication/395115183_Cognitive_Security_The_study_and_practice_of_protecting_the_human_mind_and_other_Cognitive_Assets_from_cognitive_threats
172. Clinical Surgery in General | PDF | Blood Pressure | Breathing - Scribd https://www.scribd.com/doc/5511406/Clinical-Surgery-in-General
173. Findings of the Association for Computational Linguistics: EMNLP ... https://aclanthology.org/volumes/2025.findings-emnlp/
174. [PDF] FEVER 2024 The Seventh Fact Extraction and VERification ... https://aclanthology.org/2024.fever-1.pdf

175. Dictionary of Acronyms and Technical Abbreviations - Springer Nature https://link.springer.com/content/pdf/10.1007/978-1-4471-0263-2.pdf

176. Technical Reports | Department of Computer Science, Columbia ... https://www.cs.columbia.edu/technical-reports/

177. Computer AidedComputer Aided Verification Verification - Springer https://link.springer.com/content/pdf/10.1007/978-3-031-98682-6.pdf

178. Theses - NYU Computer Science Department https://cs.nyu.edu/dynamic/reports/?type=PhD&year=all

179. Arxiv今日论文| 2026-01-28 - 闲记算法 http://lonepatient.top/2026/01/28/arxiv_papers_2026-01-28

180. Sensors, Volume 18, Issue 12 (December 2018) – 426 articles - MDPI https://www.mdpi.com/1424-8220/18/12

181. multi-page.txt - World Bank Documents & Reports https://documents1.worldbank.org/curated/en/268221468051264089/txt/multi-page.txt

182. Download book PDF - Springer https://link.springer.com/content/pdf/10.1007/978-3-540-49645-8.pdf

183. Untitled - Springer Link https://link.springer.com/content/pdf/10.1007/978-0-387-73655-6.pdf

184. cmnt_vocab.txt - CMU School of Computer Science https://www.cs.cmu.edu/~ark/blog-data/data/blog_data_v1_0/dk/hbc_data/data/cmnt_vocab.txt

185. [PDF] shaping a rights‐oriented digital transformation | oecd https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/06/shaping-a-rights-oriented-digital-transformation_30378a18/86ee84e2-en.pdf

186. Global Trends in Education: Artificial Intelligence, Postplagiarism ... https://link.springer.com/article/10.1007/s40979-025-00187-6

187. Global toolkit on AI and the rule of law for the judiciary https://unesdoc.unesco.org/ark:/48223/pf0000387331

188. Defining AI Models and AI Systems: A Framework to Resolve the ... https://arxiv.org/html/2603.10023v1

189. [PDF] Using AI as a weapon of repression and its impact on human rights https://www.europarl.europa.eu/RegData/etudes/IDAN/2024/754450/EXPO_IDA(2024)754450_EN.pdf